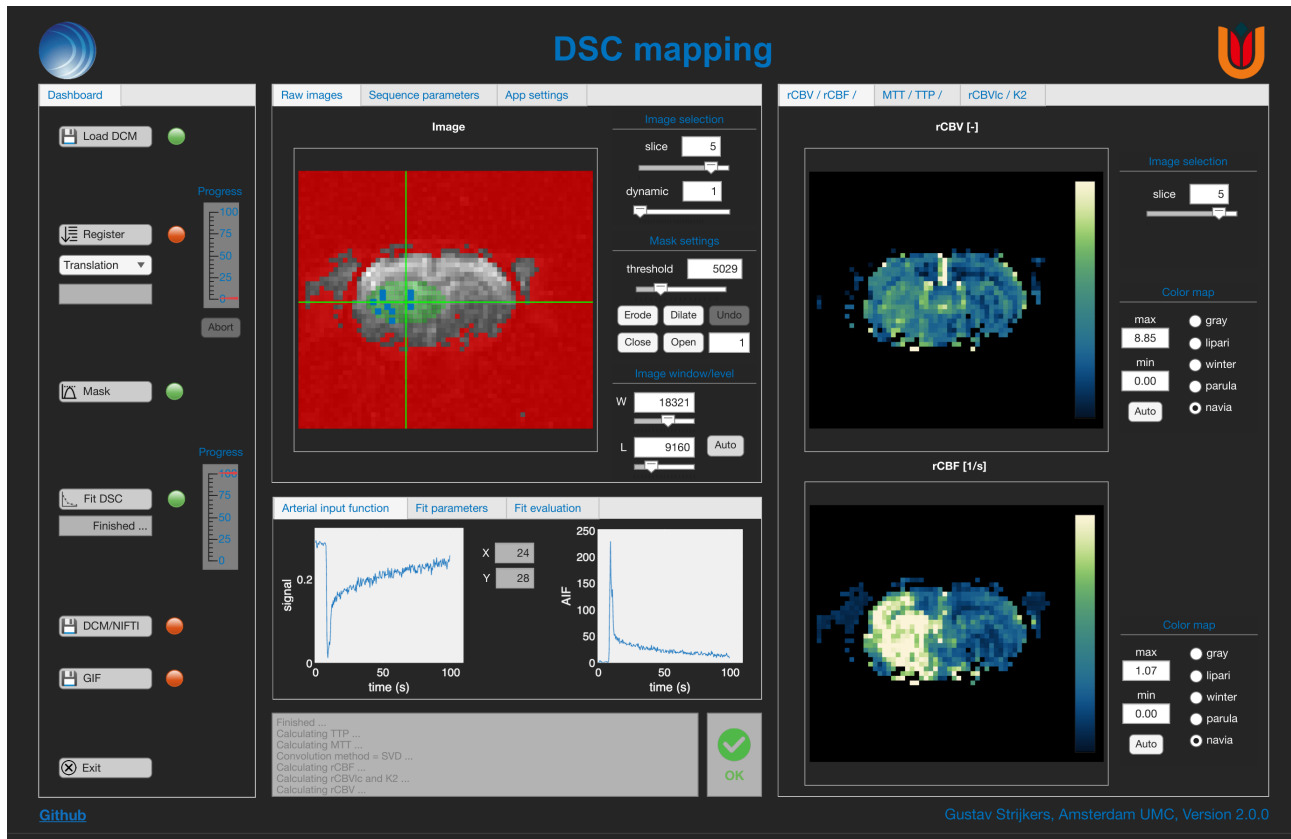




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### 1. Overview

DSCmapp is a software utility and graphical user interface to map hemodynamic and vascular properties of brain tissue from a Dynamic Susceptibility Contrast (DSC) MRI acquisition on an MR Solutions preclinical MRI system.

### 2. Contact

DSCmapp and this manual are written by:

Gustav Strijkers  
Amsterdam UMC  
Biomedical Engineering and Physics  
Amsterdam, the Netherlands  
[g.j.striekers@amsterdamumc.nl](mailto:g.j.striekers@amsterdamumc.nl)

### 3. Disclaimer

The software has been tested using mouse and rat data acquired with an MR Solutions preclinical 7.0T/24cm system.

However, this does not warrant the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free.

In case of questions or issues, please contact Gustav Strijkers.



### 4. Installation notes

#### Software download

The Matlab app and a Windows standalone version (using the free Matlab runtime engine) can be downloaded from GitHub:

<https://github.com/Moby1971?tab=repositories>

#### Installation of the Windows standalone version

DSCmapp\_app\_installer.exe

Will install the Matlab runtime engine and the DSCmapp program.

#### Elastix registration

Image registration uses elastix, which comes with DSCmapp for macOS, Windows and Linux. The macOS version is an Intel program: on a Mac with Apple silicon it runs only when Rosetta 2 is installed. When elastix cannot be run, DSCmapp registers the images with Matlab's imregtform instead, and says so at start.



## 5. Running the software

### Running in Matlab 2025a

The DSCmapp software can be started from its root directory from the command line.

```
>> DSCmapp
```

Additional licenses may be required.

```
MATLAB  
Curve Fitting Toolbox  
Image Processing Toolbox  
Optimization Toolbox  
Statistics and Machine Learning Toolbox
```

### Running the Windows standalone

The Windows standalone version can be run from the start menu or the desktop icon.

### Settings and log files

DSCmapp keeps its settings in a file, dscmappSettings.json, in a folder of its own for every version of the app:

```
macOS      ~/Library/Application Support/DSCmapp/<version>  
Windows    %APPDATA%\DSCmapp\<version>  
Linux      ~/.config/DSCmapp/<version>
```

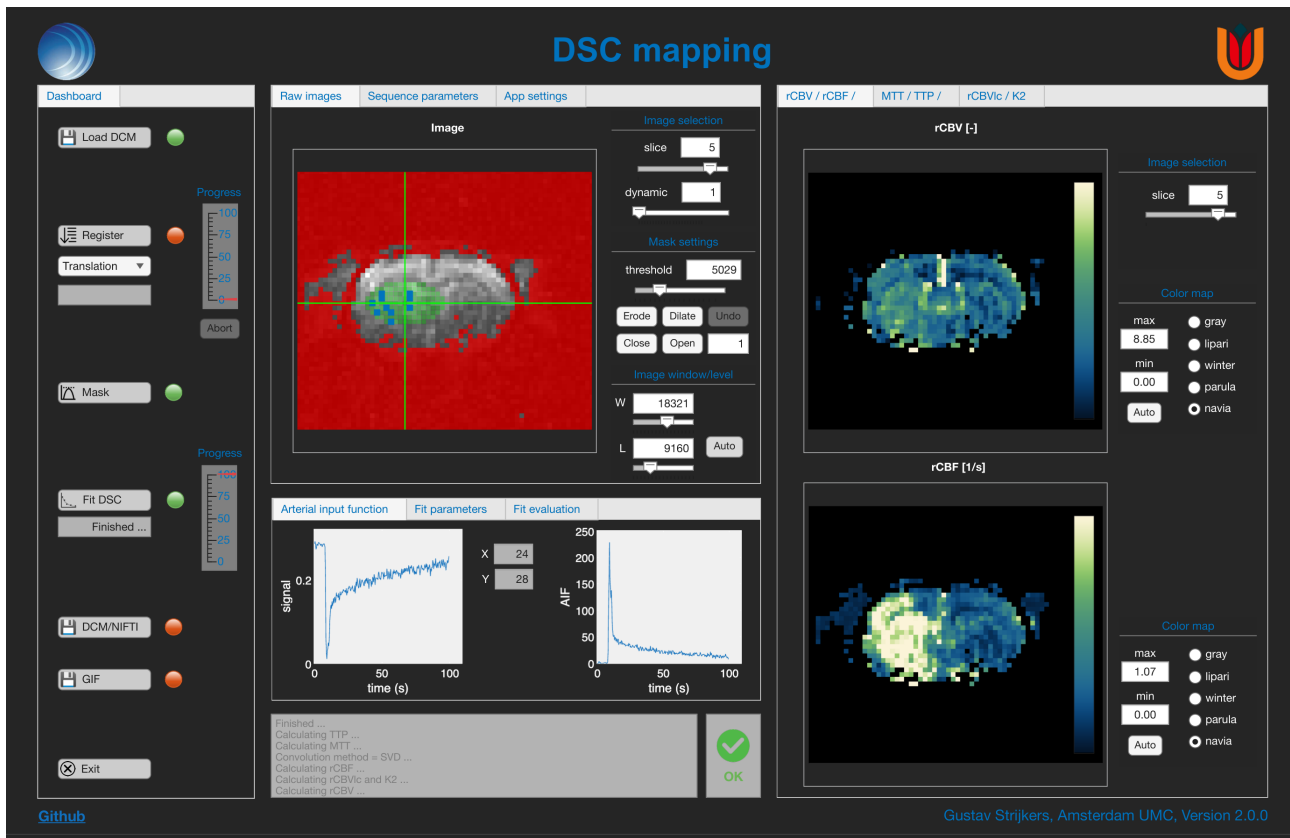
The file is written with the standard values the first time a version starts, and can then be edited by hand; changes take effect at the next start. It holds whether the app looks for a newer version, the folders where loading and the exports start, the color scheme, the sounds, the registration method, the fit parameters and the color maps. A value that cannot be used is replaced by its standard value, and the message box says so. The color scheme and the sounds can also be switched in the App settings tab, for the session only.

Every session writes a log file, dscmapp-log-<date and time>.txt, to the same folder. The log keeps every message and the parameters of each registration and fit. Logs older than the newest 50 are removed at start.



## 6. Basic operation

The DSCmapp program operates from a single window with 5 panels.



### Panel 1: Dashboard

This panel holds the task buttons, with the progress gauges, times and Abort button of registration and fit. A green light shows a task done, yellow a task running, and red one not done or failed.

### Panel 2: Raw images, Sequence parameters, and App settings

This panel has 3 tabs for the raw images, list of sequence parameters, and application settings.

### Panel 3: Arterial input function, Fit parameters, and Fit evaluation

This panel has 3 tabs: arterial input function (AIF), fit parameters, and fit evaluation.

### Panel 4: rCBV/rCBF/MTT/TTP/rCBVlc/K2

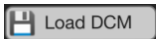
Calculated parameter maps, in 3 tabs of 2 maps each.

### Panel 5: Messages

Displays program messages and the status: OK, BUSY, WARNING or ERROR.



## Step 1: Loading data

Press  to load the DICOM files: choose the folder that holds the series' .dcm files (.../DICOM/<series>/1 in the scanner's export).

The app expects multi- or single-slice brain data with at least 50 dynamics (time series), acquired during injection of a contrast agent. The files are read in name order and checked: every dynamic has to hold the slice positions of the first. A series that does not pass is refused, with the reason in the message box.

Relevant acquisition parameters will be shown in panel 2.

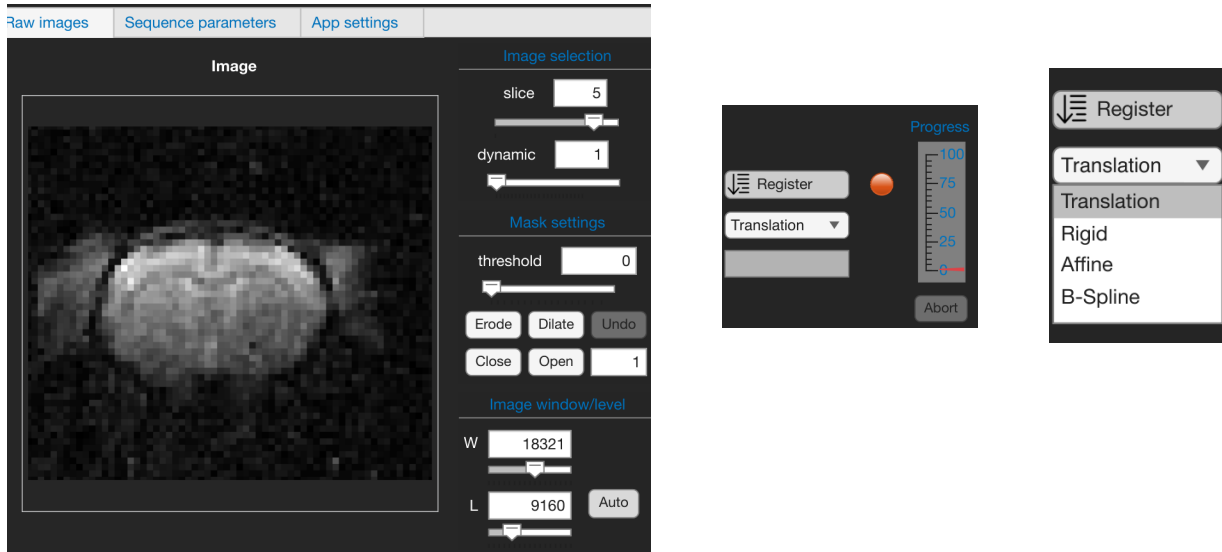
| Raw images                                                                                 | Sequence parameters                | App settings         |                                   |
|--------------------------------------------------------------------------------------------|------------------------------------|----------------------|-----------------------------------|
| DCM folder <input type="text" value="/Users/gustav/MRI-data/DSC/Data/1111/DICOM/23574/1"/> |                                    |                      |                                   |
| TR (ms)                                                                                    | <input type="text" value="350"/>   | FOV (mm)             | <input type="text" value="25.6"/> |
| TE (ms)                                                                                    | <input type="text" value="15"/>    | dim X                | <input type="text" value="64"/>   |
| flip angle                                                                                 | <input type="text" value="25"/>    | dim Y                | <input type="text" value="56"/>   |
| #dynamics                                                                                  | <input type="text" value="285"/>   | #slices              | <input type="text" value="6"/>    |
| aspect ratio                                                                               | <input type="text" value="1.143"/> | slice thickness (mm) | <input type="text" value="1.4"/>  |

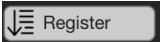
By default, the first 14 dynamics are left out, to remove the transient signal during the acquisition of the first dynamics: the analysis starts at dynamic 15. If needed, this can be changed with First dynamic in the Fit parameters tab, which clears a registration, the mask, the AIF and the maps.

| Arterial input function   | Fit parameters                    | Fit evaluation |
|---------------------------|-----------------------------------|----------------|
| <b>Data selection</b>     |                                   |                |
| First dynamic             | <input type="text" value="15"/>   |                |
| <b>AIF settings</b>       |                                   |                |
| ROI y axis (%)            | <input type="text" value="15"/>   |                |
| ROI x axis (%)            | <input type="text" value="15"/>   |                |
| Max # AIF pixels          | <input type="text" value="10"/>   |                |
| Quadratic R2* [CA]        | <input type="checkbox"/>          |                |
| <b>DSC fit parameters</b> |                                   |                |
| SVD                       | <input type="radio"/>             |                |
| cSVD                      | <input type="radio"/>             |                |
| oSVD                      | <input type="radio"/>             |                |
| Kh                        | <input type="text" value="1.00"/> |                |
| $\rho$                    | <input type="text" value="1.00"/> |                |

## Step 2: Image registration

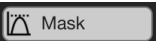
Use the sliders and/or edit-fields to inspect the slices and dynamics. The image scale can be adjusted with the Window (W) and Level (L) edit fields and sliders.



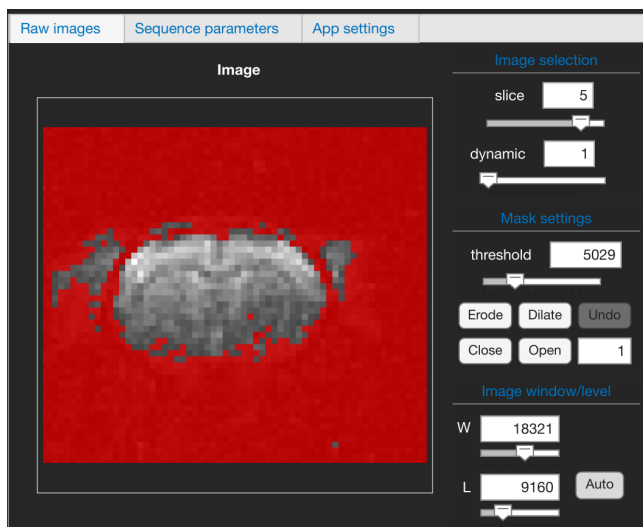
If the images are not perfectly registered, press  to register them with elastix, as a translation, rigid, affine or B-spline transform; Abort stops it.

NOTE: usually not needed, slow for large series, and it clears the mask, AIF and maps.

## Step 3: Segmenting the images

To isolate the brain the images can be segmented. Press  for an automatic threshold (Otsu) per slice.

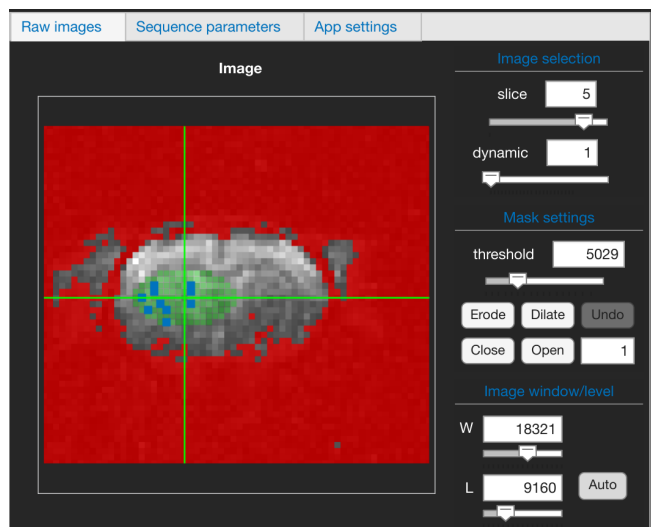
The result will look something like the figure below in which the red pixels indicate the regions that will be set to zero in the DSC parameter maps.



If the automatic threshold is not optimal, it can be adjusted per slice. The segmentation applies to all dynamics and can be refined by erode, dilate, close and open operations on the red region, of the size beside them; Undo takes the last one back.

## Step 4: Selecting an arterial input function (AIF)

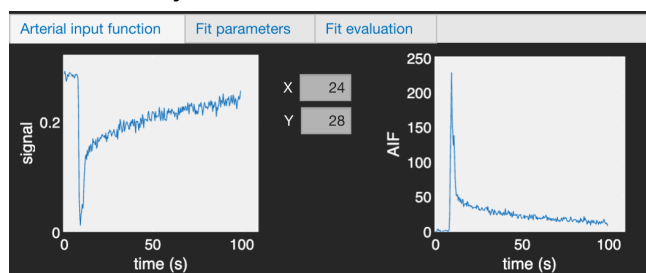
To semi-automatically determine an arterial input function, mouse-click on the brain in the image. The AIF is searched in the slice shown, around the voxel clicked.



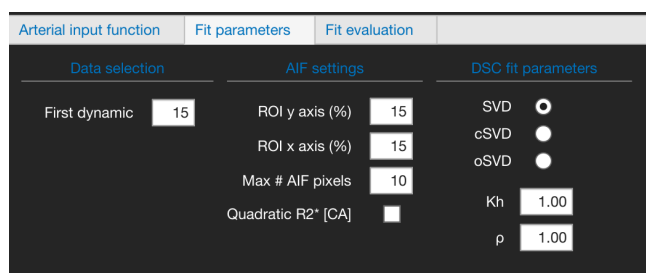
To determine the correct AIF the following approach should be considered. Make a selection for which the signal change is early (first to respond), deep (substantial signal loss), and narrow. Selecting a voxel within an artery might not always be advisable because of signal dephasing by a high contrast agent concentration in the artery. It is acceptable to select voxels adjacent to the artery as long as the local signal drop is early, deep and narrow.

The AIF is automatically selected from the green-shaded area following a set of criteria (based on: Peruzzo et al., Comput Methods Programs Biomed. 2011;104(3):e148–57).

The AIF may look like this; when none is found, the AIF found before is kept.



The size of the green-shaded AIF search area and the maximum number of selected AIF pixels can be adjusted in the Fit parameters tab.



## Step 5: Fitting the hemodynamic parameters

The app estimates the CBF by solving the following convolution equation:

$$c(t) = CBF \cdot AIF(t) \otimes R(t)$$

with:

$c(t)$  = the tissue contrast concentration

$AIF(t)$  = the arterial input function

$R(t)$  = the fraction of contrast agent present in the volume of interest (residue function)

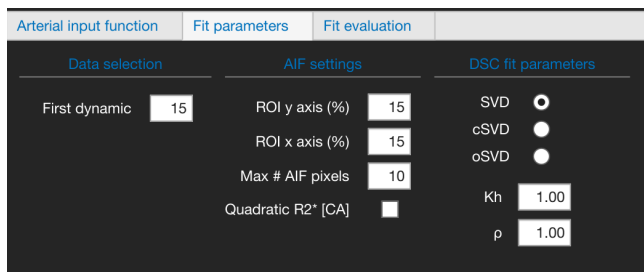
The app uses a model-independent singular value decomposition (SVD) deconvolution approach and makes no assumptions on the CBF and on the residue function shape, thus treating them as complete unknowns to be estimated from the data.

SVD is based on the decomposition of a matrix AIF to compute the inverse  $AIF^{-1}$  and then derive the residue function and the CBF.

Three types of SVD can be chosen in the Fit parameters tab: standard SVD, circular Singular Value Decomposition (cSVD), and oscillation index Singular Value Decomposition (oSVD).

For the AIF, a linear (not checked) or quadratic (checked) dependency of  $R2^*$  on the contrast agent concentration can be chosen.

For quantitative perfusion values one can adjust  $\rho$ , which is the density of brain tissue (typically, 1 g/ml tissue) and  $K_h$  which is the hematocrit factor (typically, 0.67 in the capillaries).



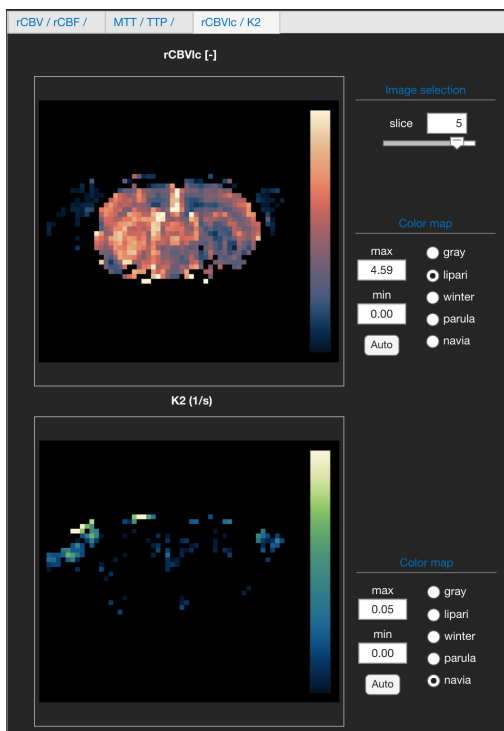
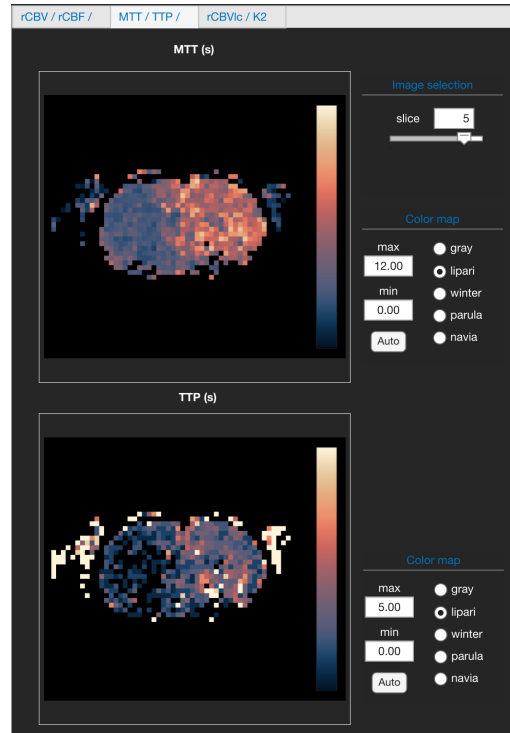
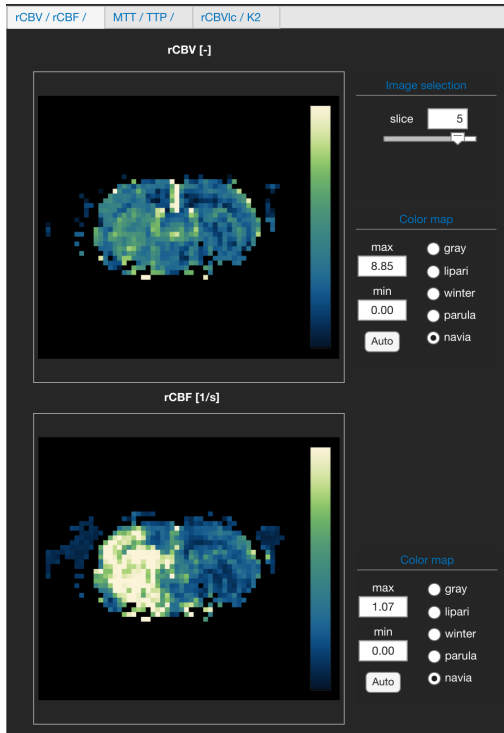
To start the fitting, press  . The AIF is searched again at the last click.

The application will produce the following maps:

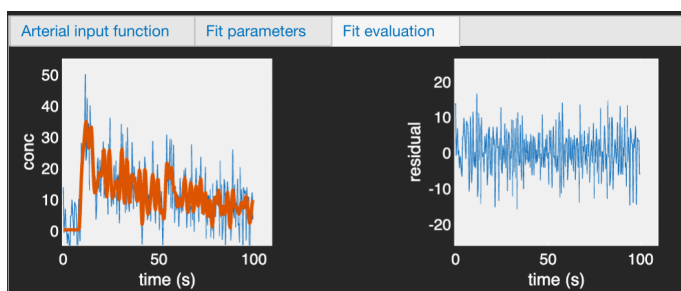
|               |   |                                                          |
|---------------|---|----------------------------------------------------------|
| <b>rCBV</b>   | = | relative CBV [-]                                         |
| <b>rCBF</b>   | = | relative cerebral blood flow [1/s]                       |
| <b>MTT</b>    | = | mean transit time (s)                                    |
| <b>TTP</b>    | = | time to peak / delay between AIF and tissue response (s) |
| <b>rCBVlc</b> | = | relative CBV with leakage correction [-]                 |
| <b>K2</b>     | = | leakage parameter (1/s)                                  |

## Step 6: Inspecting the results

The hemodynamic parameter maps will be shown in the panel on the right. Each map has its own color map (gray, lipari, winter, parula or navia); the slice fields of the three tabs move together, and the mouse wheel over the maps steps through the slices.



By clicking on the raw image of panel 2 after the fit, the Fit evaluation tab shows the data (blue line) and fit (orange line) of that voxel, and on the right their difference. The click also searches an AIF around the voxel, shown in the Arterial input function tab; the maps keep the AIF they were fitted with. Each map has a max and a min for its color scale, which Auto sets to 3 times the median of the slice shown and to 0.



## Step 7: Exporting the maps

There are two ways to export maps.

### (1) DCM/NiftI



Exports the six maps, masked, in DICOM and NiftI format for further processing in 3rd party software. Each DICOM map is a series of its own.

### (2) GIF



Exports the six maps as animated GIF files, one frame per slice, in the color maps and scales shown, with two pictures of the app's window.

When the DICOM files were loaded from the scanner's folders, the exports go beside them, in .../Data/<study>/DICOM, Nifti and GIF; otherwise the app asks for a folder. Every export gets a new numbered folder.

## Step 8: Exit

Press  to exit the program.



### Credits

#### Code

- The app uses code from this DSC MRI toolbox <https://github.com/FAIR-Unipd/dsc-mri-toolbox>.

#### AIF selection

- Peruzzo et al. Automatic selection of arterial input function on dynamic contrast-enhanced MR images. *Comput Methods Programs Biomed.* 2011;104(3):e148–57.
- Calamante et al. Arterial input function in perfusion MRI: A comprehensive review. *Prog Nucl Magn Reson Spectrosc.* 2013;74:1–32.

#### Vascular permeability

- Skinner et al. On the Use of DSC-MRI for Measuring Vascular Permeability. *Am J Neuroradiol.* 2016;37(1):80–7.

#### CBF calculations

- Peruzzo et al. Stable spline deconvolution for dynamic susceptibility contrast MRI. *Magn Reson Med.* 2017;78(5):1801–11.
- Zanderigo et al. Nonlinear Stochastic Regularization to Characterize Tissue Residue Function in Bolus-Tracking MRI: Assessment and Comparison With SVD, Block-Circulant SVD, and Tikhonov. *IEEE Trans Biomed Eng.* 2009;56(5):1287–97.

#### AIF correction for quadratic $R2^*$ dependency on contrast agent concentration

- van Dorth et al. Dependency of  $R2$  and  $R2^*$  relaxation on Gd-DTPA concentration in arterial blood: Influence of hematocrit and magnetic field strength. *NMR Biomed.* 35, e4653 (2022).